Rebuild around the project

Study findings for a construction ERP covering tendering through handover and snagging.

The strongest confirmed problem is the organisation and continuity of work. The ERP contains substantial construction functionality, but users must navigate module names, transaction lists, generic project forms, and several dashboard layers to reach it.

A new application should make the project, work package, location, current owner, and next action clear throughout the lifecycle. Rebuilding the same menu structure in new technology would preserve much of the present difficulty.

What the live walkthrough established

Confirmed finding	Implication for the rebuild
42 primary administrator entries; 267 leaf menu controls	Reduce primary choices and organise daily work by role and project context.
Project entry opens on a map; key links are inside More	Make current work and exceptions the project landing content.
Requisition, PO, and receipt are linked, but context fades	Keep a visible project / package / location context across separate records.
A flat 302 snag is structurally located only at Main site	Validate floor, unit, zone, and system relationships for tower operations.
Variation and closeout samples focus on commercial value	Validate instruction/time linkage and technical handover separately.

What is already useful

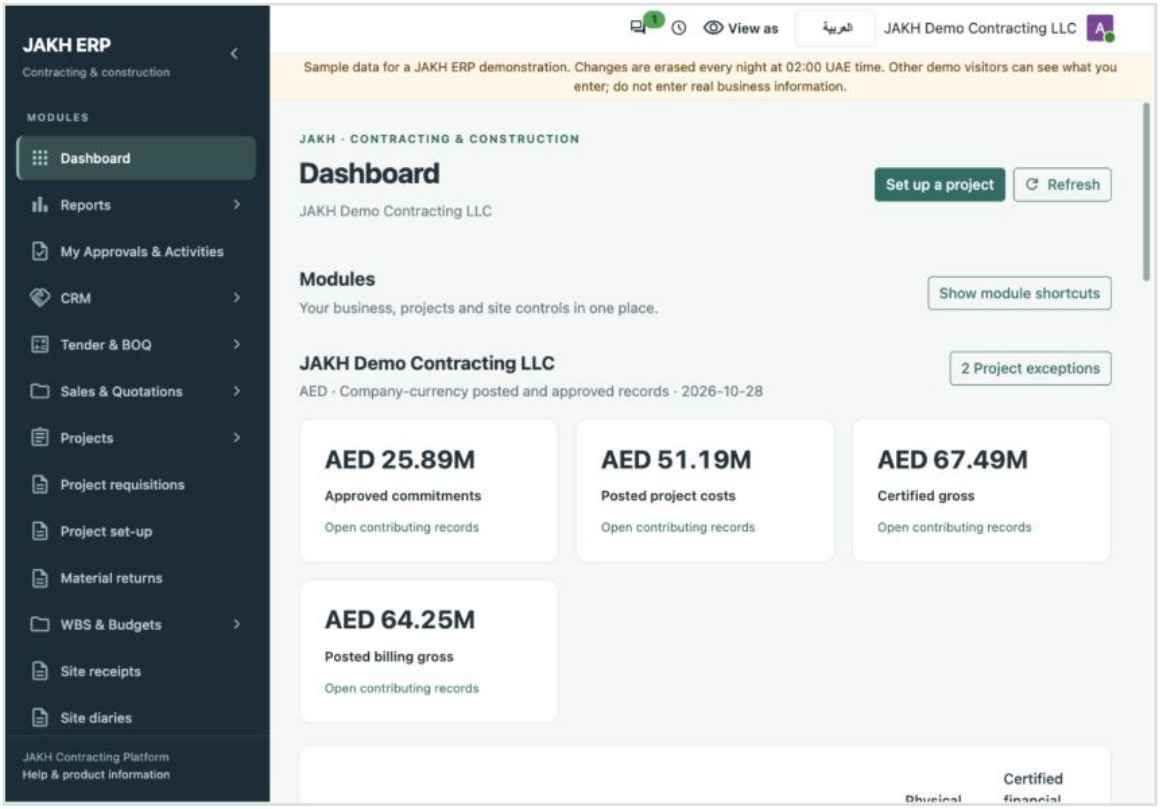
Role filtering, linked records, current/superseded drawing revisions, textual statuses, cumulative measurements, and approval history are visible. These business capabilities should survive the simplification.

Boundary: this is an expert read-only UI walkthrough and a proposed rebuild brief. It does not prove full functional coverage, accounting correctness, accessibility compliance, security, performance, or usability with actual staff. No application code was started.

Navigation needs a new structure

Evidence: E39 administrator dashboard, E41 expanded navigation inventory, E30/E50/E53 role dashboards.

Role	Primary entries	What this proves
Administrator	42	Broad modules and individual transactions share one level.
Project Manager	20	Role filtering still leaves a long operational menu.
Site Engineer	10	A smaller role-specific menu already exists.
Estimator	9	Tender work can be presented with fewer primary areas.



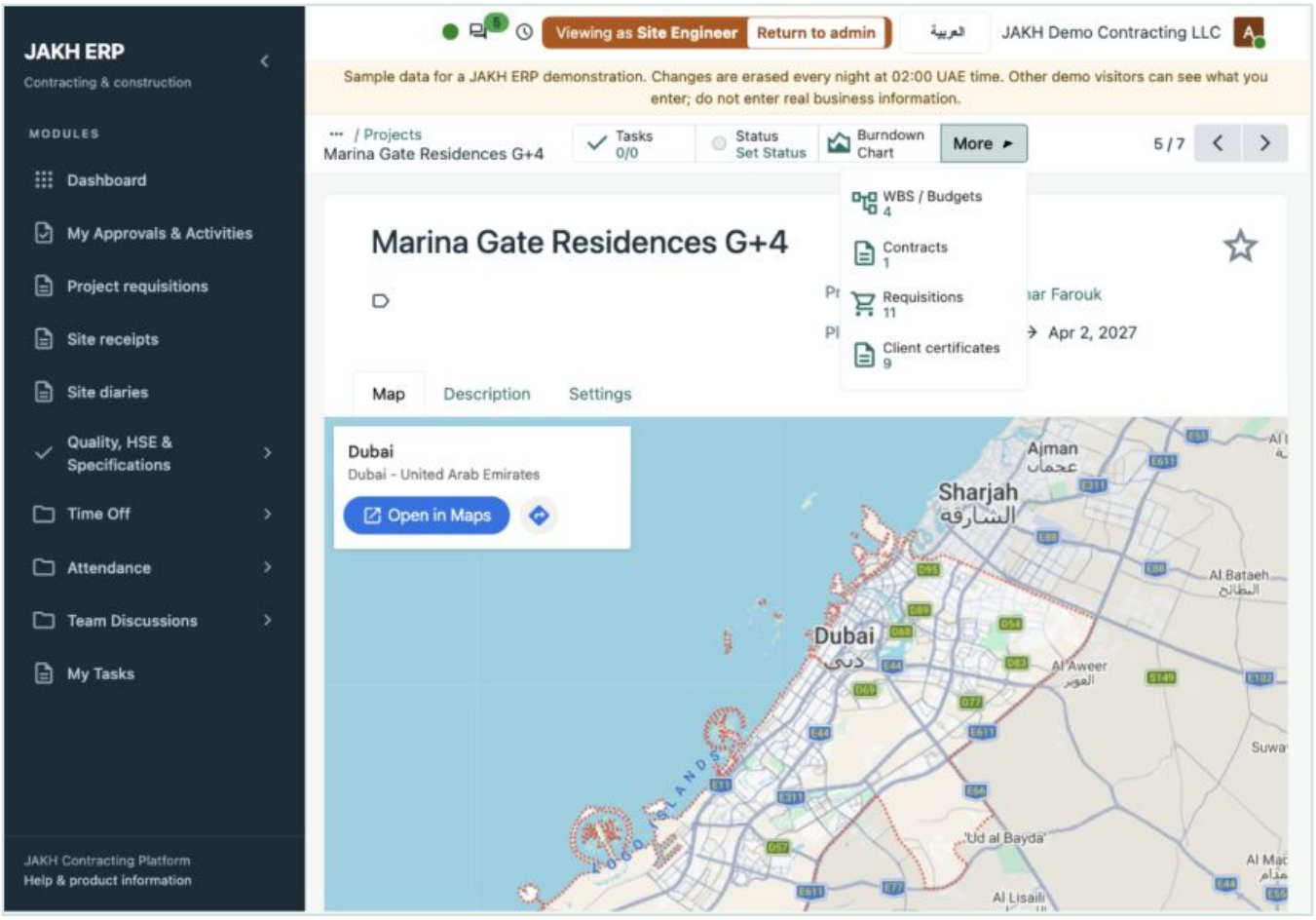
E39. Administrator in JAKH Demo Contracting LLC. The screenshot shows the first portion of the scrollable sidebar.

The fully expanded administrator tree contains **363 button controls: 96 groups and 267 leaf buttons**, with paths reaching four menu levels. Leaf counts are navigation controls, not a count of unique features.

Repeated leaf labels include Products (6), Customers (4), Settings (12), Dashboard (4), Payments (3), and My Tasks (2). Their targets were not all compared; identical labels do not prove duplicate records or duplicate functionality.

A project should open on work

Evidence: E32 project entry and E33 its More menu, viewed as Site Engineer.



E33. Marina Gate Residences G+4 opens on Map / Description / Settings. WBS, Contracts, Requisitions and Client certificates sit under More.

Observed friction

The main project view prioritises a large Dubai map, generic task counts, and project metadata. It does not immediately answer what is happening on site, what is late, what needs approval, or which floor/package needs attention.

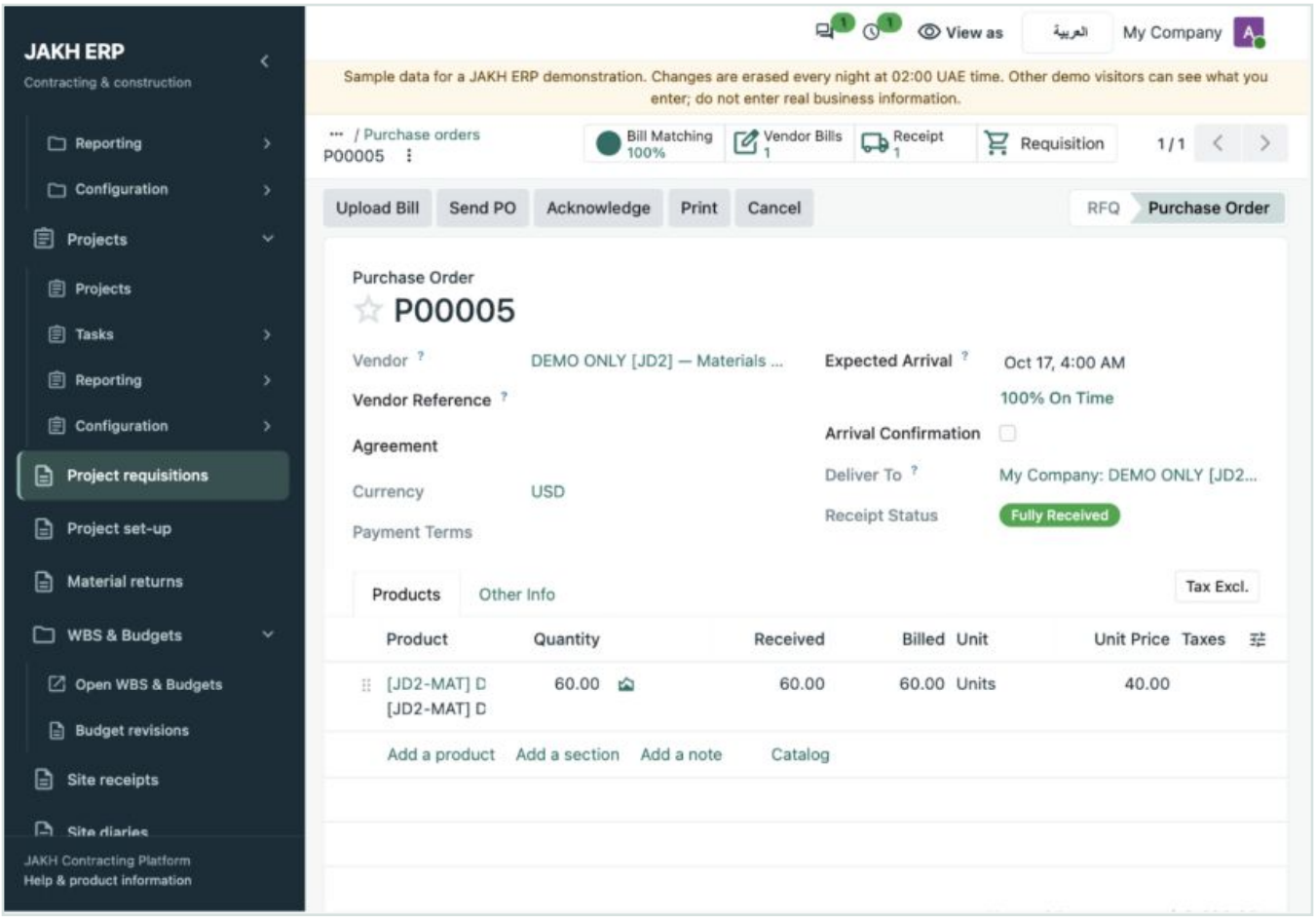
Requirement to validate

Use a project workspace with a stable identity and location/package filter. Put current actions, blockers, recent progress, and relevant documents within that context. Treat maps and administrative settings as supporting information.

Waiting records should name the next owner and expected review date, including when the viewing user cannot approve. The absence of an approval button for a site role may be correct permission behaviour.

Keep the procurement chain connected

Observed path: requisition -> linked PO list -> purchase order -> completed receipt. Evidence E19-E22.



E21. P00005 shows received/billed quantities and links to the receipt, bill, and requisition. Its main view does not give project/WBS/location the prominence seen on the requisition.

What works, and where context weakens

The requisition includes the project, WBS, required date, planning-demand reference, and requested location. The PO exposes the vendor, arrival, receipt status and quantities. The receipt exposes the source PO and stock movement. The linkage is present; the user must reconstruct its construction context.

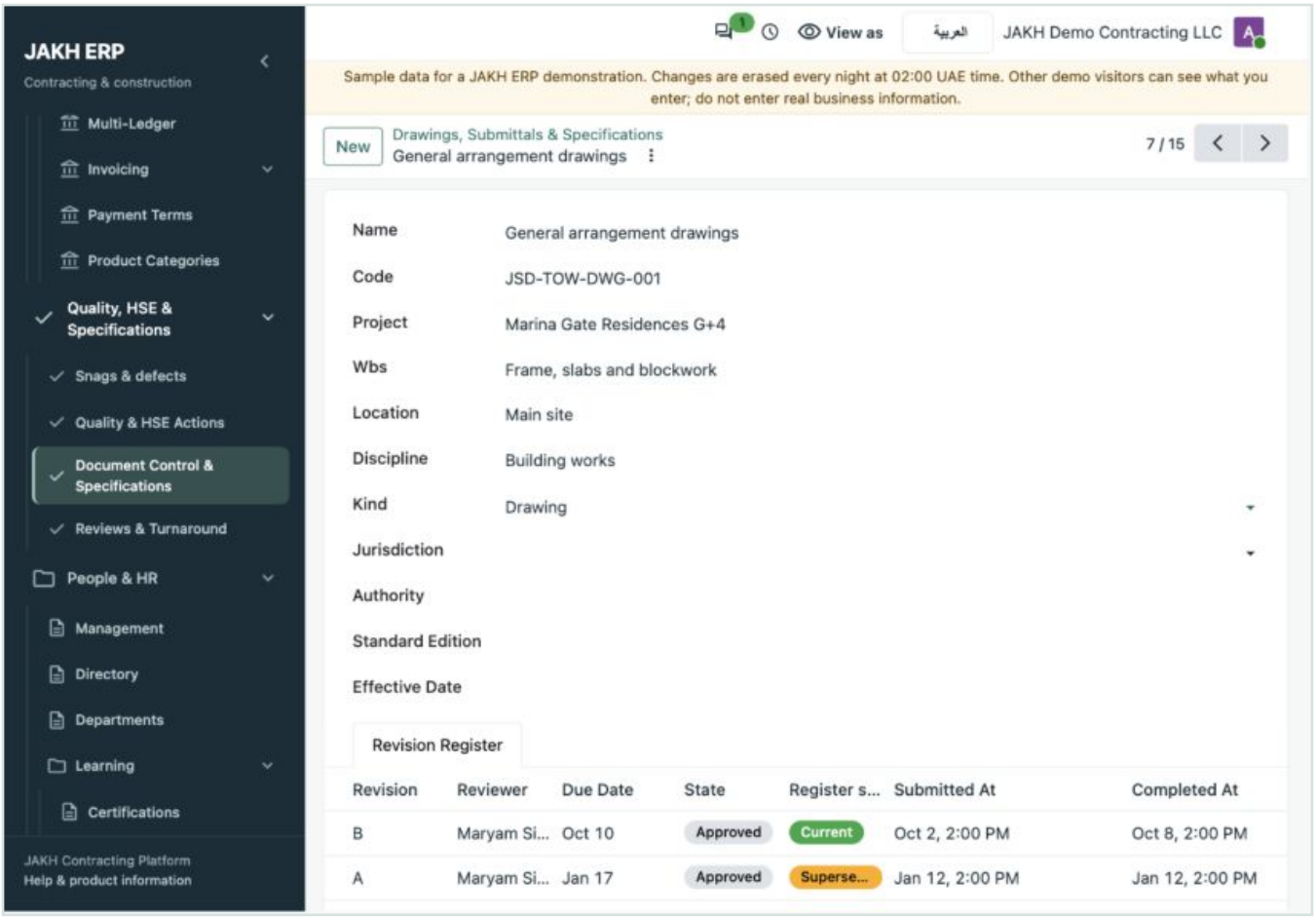
Requirement to validate

Carry project, package and location visibly across demand, request, quotation, order, receipt, inspection, issue, installation, and invoice. Link distinct controlled records rather than merging them into one editable document.

In this sample, a link labelled 1 Purchase orders opens an intermediate single-row list before the PO. Test direct opening when there is exactly one related record. Partial delivery, rejection, returns and billing exceptions were not executed.

Technical controls exist

Evidence: E43 document register, E44 revision history, E45 document-category dropdown.



E44. General arrangement drawing JSD-TOW-DWG-001 shows approved revision B as Current and approved revision A as Superseded.

Preserve revision meaning

The current-versus-superseded distinction is useful. A register also needs to let a site user find the correct approved document for the floor, zone, trade and activity without searching through unrelated projects.

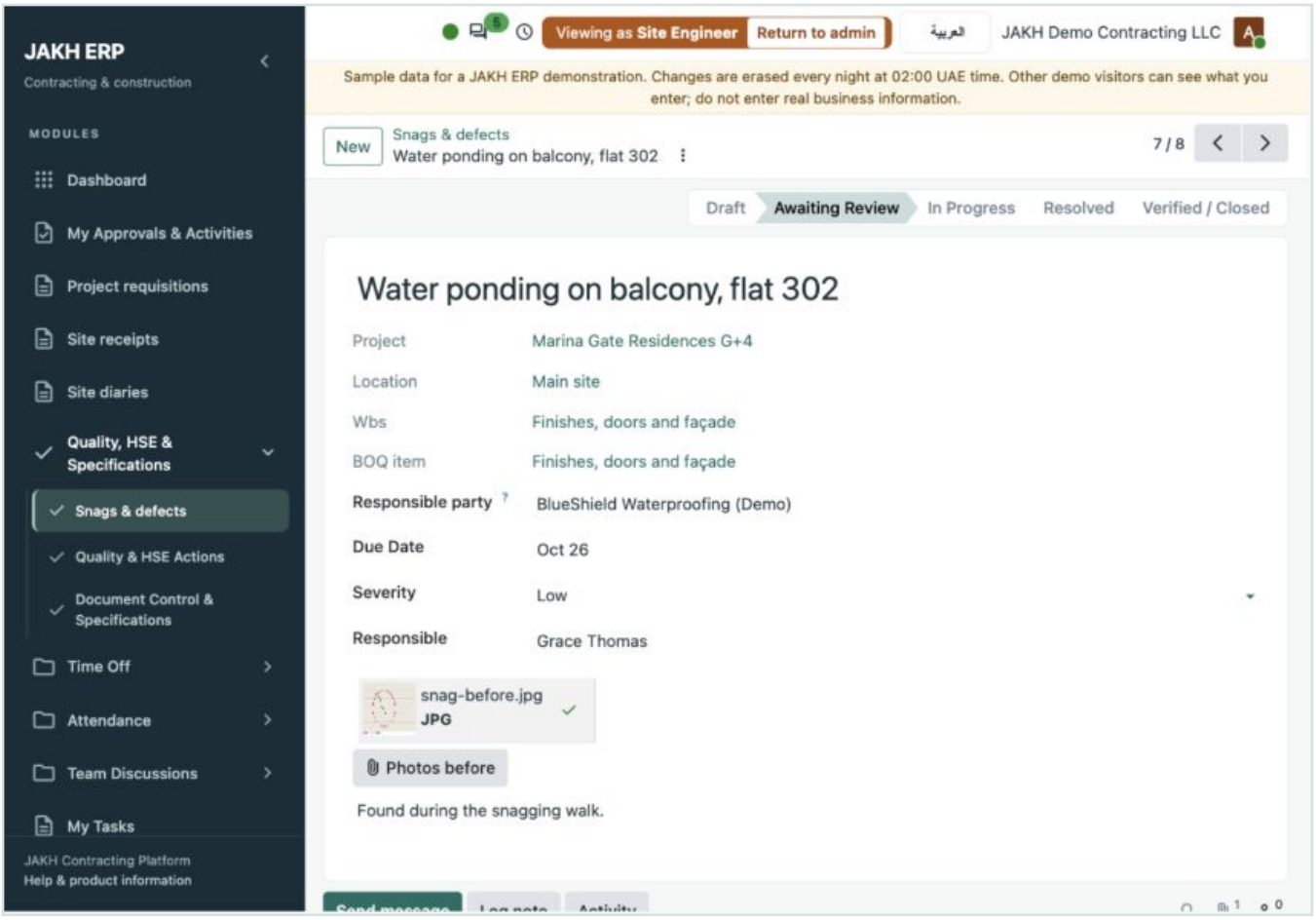
The category control offers Drawing, Submittal, Specification, Method Statement, Inspection / Test Plan, and Correspondence. The fuller register contained 15 drawing records; other category workflows were not demonstrated by populated samples.

Still to validate

Dedicated RFI routing, drawing release/transmittal, consultant response codes, inspection hold points, document distribution and external-party review. A category called Correspondence does not by itself demonstrate an RFI workflow.

Tower work needs structured locations

Evidence: E34 snag register, E35 snag detail, E37-E38 submitted site diary.



E35. Water ponding on balcony, flat 302 is linked to a project, WBS, BOQ item, contractor and person. Its structured Location is Main site.

A concrete retrieval problem

The eight displayed open snags all use Main site, including titles referring to a flat or individual villa. For these records, finding work by floor or unit depends on free text. This is a sample-data limitation and an information-structure risk; it does not prove the application cannot represent locations.

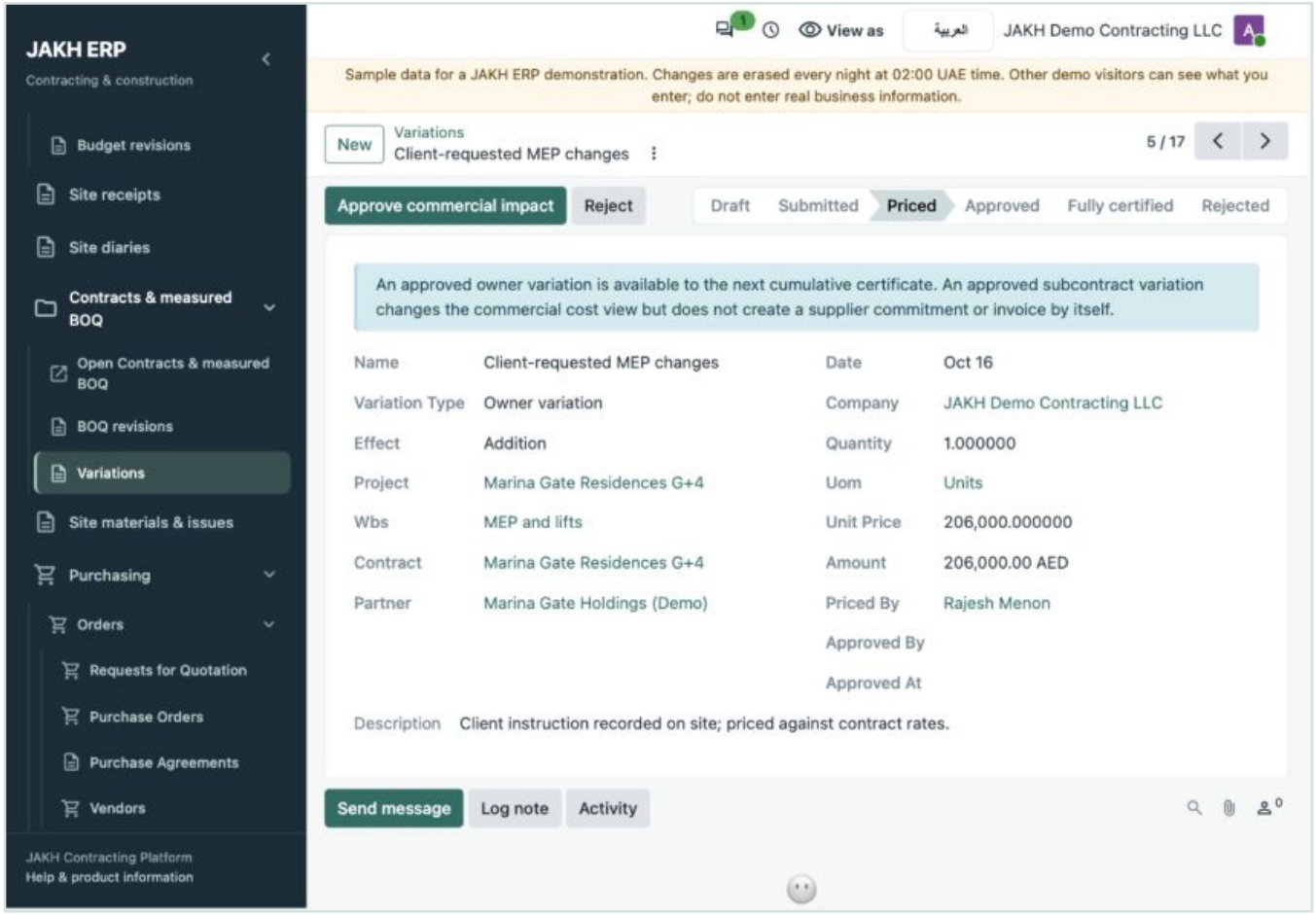
Requirement to validate

Model project -> tower/building -> floor -> unit/zone, with systems and packages linked separately. Test bulk snagging, photos, contractor assignment, rectification, reinspection, reopening and partial handover by location.

The site diary has purposeful Manpower, Equipment, Photos, and Work and events tabs. Keep useful grouping. Its Submitted state and the snag's Awaiting Review state need clearer next-review ownership in the captured views.

Connect changes to their consequences

Evidence: E46-E47 variation records; E15 client certificate and E29 subcontract service entry.



E47. A priced MEP variation has project/contract links, quantity, rate, value, pricing owner, and approval actions. No instruction link, notice deadline, or time impact is visible on this record.

Commercial coverage is present

The sampled client certificate contains cumulative quantities, prior/current/remaining measurements, approved variations, retention/recovery values, an invoice link, and technical/commercial/finance review history. A submitted subcontract service entry also has package, location and quantity context.

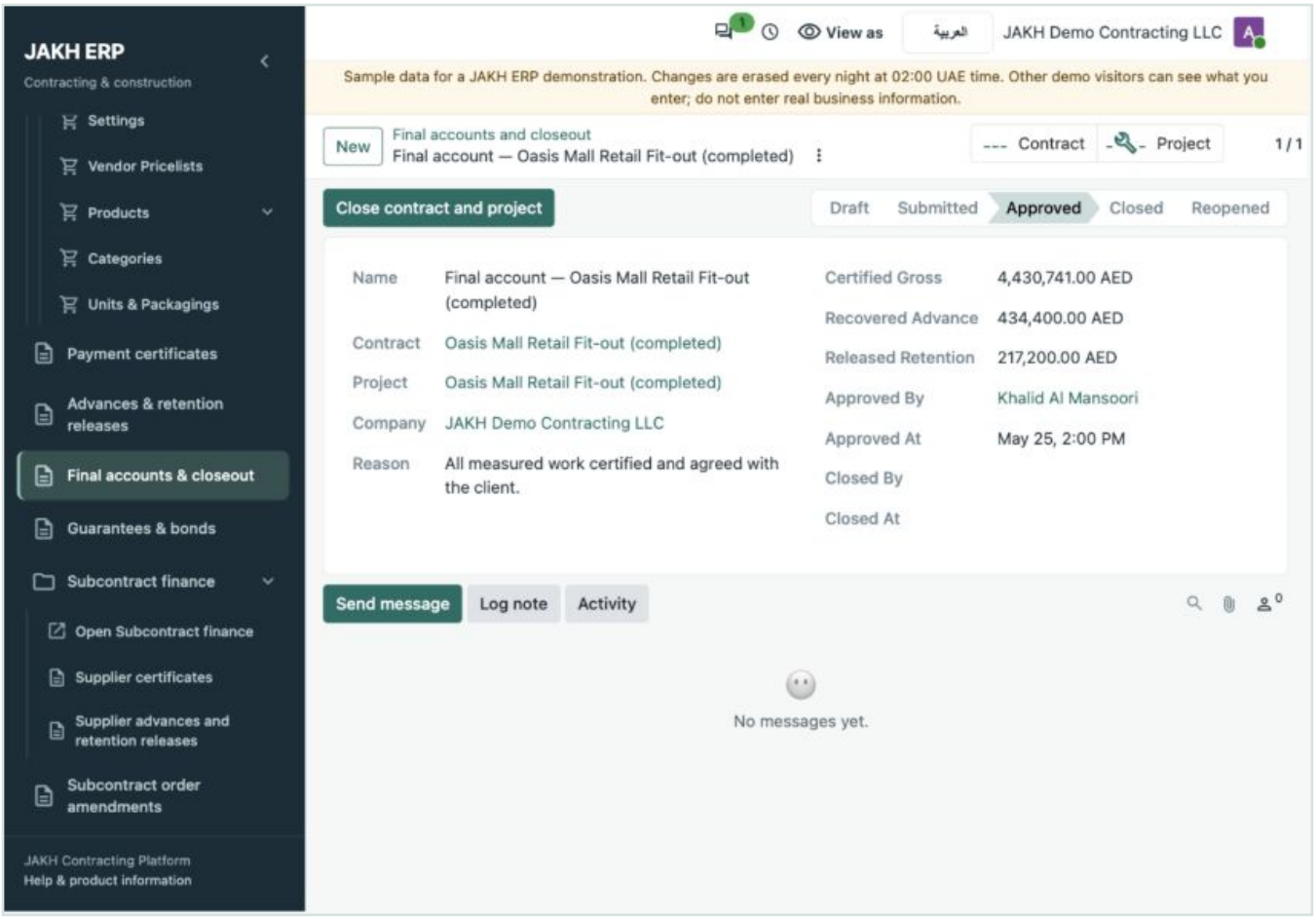
The chain to validate

Source instruction -> required notice -> estimate and cost/time impact -> review -> approval or dispute -> budget/programme change -> measured work -> certification -> invoice -> cash. Include work proceeding before approval and rejected or partially certified amounts.

Missing fields on the sampled variation are not proof of system-wide absence. Contract-specific notice rules, approval authorities and finance treatment require validation with your business owners.

Handover is more than a final account

Evidence: E48-E49 final-account workflow; E34-E35 snagging. No closure action was executed.



E49. The completed retail-fit-out project has an Approved final account and a Close contract and project action. The visible detail focuses on certification, advance recovery and retention.

Separate three completion questions

Technical readiness: are inspections, testing/commissioning, authority requirements, as-builts, O&M manuals, warranties, training and required snags complete?

Client acceptance: which building, floor, unit or system is accepted, by whom, with what outstanding obligations and defect-liability dates?

Commercial settlement: are measurements, variations, claims, subcontract settlements, retention and final balances agreed?

The sampled final-account screen does not visibly present a structured technical-readiness pack. Backend closure checks and wider handover workflows were not tested. Define the three tracks explicitly before designing the replacement.

Construction coverage map - 1

Observed means visible in this run. Not demonstrated means unverified, not proven absent. Execution/approval transitions were not performed.

Lifecycle area	Current evidence	Requirement / next verification
Tender pipeline / bid decision	Observed: generic CRM entry; estimator preview has no matching opportunities (E51).	Validate bid/no-bid, tender deadlines, qualifications, clarifications, tender security and opportunity access.
BOQ / estimate / revisions	Observed: estimates, study versions, pricing, review and transfer links (E05, E52).	Validate take-off detail, quote comparison, exclusions, alternate bids, revision comparison and approval limits.
Award / project mobilisation	Observed: estimate-to-quotation/project transfer; setup entry (E05-E07, E26).	Validate contract obligations, responsibilities, programme and budget baseline; direct-award path.
Project / tower locations	Observed: location records; fuller building sample uses Main site (E35, E44).	Validate tower/floor/unit/zone hierarchy and large-volume navigation. G+4 is not a high-rise scale test.
Budget / cost codes / forecasts	Observed: WBS release, budget and control report (E24-E25, E40).	Validate revisions, commitments, actuals, forecast ownership, approvals and source reconciliation.
Programme / look-ahead	Observed: P6 schedule progress register, 145 records, including manual updates (E42).	Validate activity logic, baselines, imports, progress evidence, look-ahead and delay/EOT linkage.
Drawings / submittals / documents	Observed: current/superseded drawings and six categories (E43-E45).	Validate submissions, transmittals, distribution, consultant response and overdue review ownership.
RFIs / instructions	Not demonstrated as a dedicated workflow in the inspected views.	Trace RFI -> response -> drawing/work instruction -> cost/time consequences. Do not treat a label as proof.
Materials / procurement	Observed: demand/request/order/receipt links (E19-E22).	Validate quote comparison, approvals, lead times, partial/rejected delivery, returns and material inspection.
Stores / site issues / installation	Navigation and demand links observed; full operational chain not walked.	Trace stock source, reservation, transfer, issue, installed quantity, wastage and cost allocation.
Subcontract award / amendments	Navigation and subcontract order/service links observed (E28-E29, E41).	Validate package scope, exclusions, measurement rules, changes, commitments and subcontract securities.
Subcontract measurement / payment	Observed: submitted/approved service entries; application entry (E27-E29).	Validate portal claim -> acceptance -> certificate/bill/payment; rejected work, deductions and correction.

Construction coverage map - 2

This map is a requirements register for the new ERP. Supporting functions remain in scope even when sampled only through navigation.

Lifecycle area	Current evidence	Requirement / next verification
Daily reports / labour / plant	Observed: populated diaries with resources and work/events (E36-E38); allocation/feed entries.	Validate who submits/approves, location/package detail, hours, equipment use and cost-feed reconciliation.
Quality / HSE / inspections	Quality/HSE navigation and cases/reminders observed (E30, E41).	Validate ITP hold points, inspection requests, NCRs, corrective actions, reinspection and safety events.
Variations / notices / delay	Observed: price/approval states; time/notice chain not visible in sample (E46-E47).	Define instruction evidence, notice dates, cost/time assessment, disputed work and linked approval history.
Client measurement / certification	Observed: cumulative BOQ/variation measurement and staged approval (E15).	Validate claimed vs certified/rejected values, evidence, corrections, prior quantities and invoice progression.
Finance / cash / accounting	Navigation, posted-cost reporting, bills/invoices and deductions observed (E15, E21, E40-E41).	Specify original ledger/posting, reconciliation, periods, tax/localisation and multi-company rules with finance.
Approvals / delegation / exceptions	Observed: role routes/history; PM activity queue has two threshold tasks (E15-E16, E54).	Verify queue coverage for all pending documents; delegation, absence cover, rejection/resubmission and independence.
Testing / commissioning	Not demonstrated as an end-to-end workflow in inspected records.	Define system/asset tests, dependencies, witnesses, results, failures/retests and readiness evidence.
Snagging / rectification	Observed: populated snags, owners, before photo and status sequence (E34-E35).	Validate after-photo requirements, independent verification, reopening, bulk capture and unit/system filtering.
Handover / defect liability	Not demonstrated as a structured handover pack in the sampled final-account view (E49).	Define partial acceptance, as-builts, O&M, warranties, training, DLP dates, outstanding work and responsibility.
Final accounts / retention	Observed: financial final-account workflow; sample Approved and Closed records (E13, E49).	Validate technical/contractual readiness gates, settlement, retention release and auditable reopening.
People / assets / shared masters	Navigation inventory only for most HR, attendance, leave, timesheets, fleet and plant functions (E41).	Confirm required depth. Include these in full ERP scope; do not silently omit them from the rebuild.
Portals / integrations / migration	Role chooser exposes client/subcontractor portals; schedule and cost feeds have entries.	Verify portal permissions, external reviews, integration ownership, data/attachment export, migration and cutover.

A simpler structure to validate

Proposed information architecture only. No visual direction, interface prototype or application code has been produced.

Test a small set of stable work areas, filtered by responsibility. The exact count and names should follow task testing rather than become an arbitrary restriction.

Candidate area	Purpose
My Work	Approvals, assigned work, blockers and overdue actions across permitted projects. Each item identifies project, owner, due date and next action.
Tenders	Opportunities, tender documents, BOQ, estimates, comparisons, revisions and the award handoff.
Projects	The project workspace: overview, programme, site delivery, engineering/quality, commercial work and handover, revealed by stage and role.
Buying	Cross-project purchasing/subcontract queues and supplier work. Project views filter the same underlying records; they do not create copies.
Finance	Project commercial-to-accounting handoffs, receivables/payables, cash, ledger and period work, with appropriate authority.
Resources	People, equipment, stores/catalogues and shared records, separated by task and role within this area.
Administration - utility access	Company setup, permissions, workflow policies and integrations, outside the daily operational menu.

Rules for keeping it simple

Keep project / package / location visible. Show the few actions valid now. Use one canonical record with contextual views. Enter shared facts once; preserve approved baselines and document snapshots with traceable revisions. Put reports where the related work happens.

Repeated BOQ views can represent estimate, awarded baseline and certified measurement - different business meanings. Remove unnecessary re-entry and competing navigation; do not erase the history or controls that make those stages trustworthy.

Define an original application

Rebuild boundary and migration study. The current product supplies observable business requirements, not implementation code.

No Odoo application-code dependency

Build original source, data models, APIs, approval/state rules and UI. Do not copy, translate or adapt Odoo modules or application code. No existing source code was inspected during this study. Selection of framework, database and deployment architecture remains a later engineering decision.

Core domain relationships to specify

Company and authority -> party/person -> opportunity/tender -> estimate revision -> contract and BOQ baseline -> project, location and WBS/package -> demand and commitment -> delivery/service/progress evidence -> variation/certificate -> invoice/payment -> handover/defect obligation.

Keep document revisions, financial postings, approval events and operational evidence linked but distinct. Define stable identifiers, permissions, immutable approved snapshots, corrections, search, attachments and audit history across those relationships.

Finance and supporting ERP scope

Specify original accounting/posting and reconciliation requirements with the finance owner. Inventory the required HR, attendance, leave, timesheet, plant, fleet, master-data and administration functions. Integration decisions require explicit scope; they must not quietly replace requested functionality.

Migration and cutover requirements

Step	Evidence required before production use
Inventory and export	Authorised exports of records, identifiers, attachments, companies, currencies, statuses, approvals and ownership.
Map and trial-migrate	A field/state map and repeatable trial import; preserve relationships and approved historical meaning.
Reconcile	Counts, opening balances, inventory, open orders, project budgets/costs, certificates, claims, retention and attachments checked by owners.
Pilot and cut over	One representative project plus exceptions; parallel reconciliation, backups, rollback and agreed transition responsibilities.

Validate before detailed design

The read-only study is ready for review. Detailed design and application coding remain behind the requested review point.

Priority sequence - the full scope remains

Priority	Outcome
First: navigation and context	Agree roles, canonical records, project/floor/package context, terminology and an understandable work queue.
Next: tower workflow depth	Resolve engineering-to-site, instruction-to-variation/time, package-to-measurement, and technical-to-commercial closeout handoffs.
Then: coherent original ERP	Design and build the lifecycle in increments, including original finance controls and the required supporting modules.
Before replacement	Trial migration, representative-user validation, accounting/control review, security/access testing, integration checks and a controlled pilot.

Tasks for the next validation round

1. Find the approved drawing for one tower floor and confirm which revision may be used.
2. Request a material, follow a partial/rejected delivery, and find the installed quantity.
3. Trace a subcontract package from measured work through payment.
4. Record an instruction, identify cost/time consequences, and follow its approval or dispute.
5. Explain claimed, certified, invoiced and collected values for one period.
6. Find all snags in one unit, verify the fix, and check partial-handover readiness.

Use you and representative estimator, site, procurement, QS/PM, quality/document-control and finance users where available. Measure unassisted completion, time to the correct information, wrong turns, repeated entry and missed handoffs. Set numerical targets after obtaining a real baseline; no staff usability test has yet been run.

Decisions still needed

Confirm actual roles/approval authority, contract types, floor/unit structure, consultant/client/subcontractor participation, document and programme tools, accounting/localisation needs, languages/mobile/offline needs, project volumes and migration depth. A delivery estimate should follow this scope, not precede it.

Terms used in the study

BOQ: bill of quantities. WBS: work breakdown structure. QS: quantity surveyor. RFI: request for information. ITP: inspection/test plan. EOT: extension of time. DLP: defect-liability period. O&M: operation and maintenance.

Evidence and limits

Primary source: live UI at https://demo.jakh.net/ on 28 October 2026. Evidence IDs match saved screenshot and accessibility-snapshot prefixes.

Evidence	Observed route / role / purpose
E02, E39, E41	Administrator dashboards and complete expanded sidebar. E41 is an accessibility inventory, not a single screenshot.
E05-E07	My Company / USD: estimate EST/2026/00004 -> linked project and project shortcuts.
E15-E17	My Company / USD: measured client certificate -> contract -> cash/closeout tab.
E19-E22	My Company / USD: material-shortage requisition -> P00005 -> JD2/IN/00001 receipt.
E28-E29	My Company / USD: subcontract service entries and submitted review record.
E30-E38	Site Engineer, JAKH Demo Contracting LLC / AED: project portfolio, G+4 project, snags and diary.
E40, E42	Administrator, fuller company: G+4 construction report and schedule-progress register.
E43-E45	Administrator, fuller company: document register, G+4 drawing revision history and category menu.
E46-E49	Administrator, fuller company: variations and the approved completed-project final account.
E50-E52	Estimator: role dashboard, CRM entry and eight BOQ studies.
E53-E54	Project Manager: role dashboard and two-item activities/approvals queue.
E55	Administrator returned to original My Company context and normal browser viewport.

Scope limits that affect the conclusions

The first administrator view selected My Company (USD); role previews opened JAKH Demo Contracting LLC (AED), which has fuller samples. Empty initial lists are not proof of absent features. The study inspected a G+4 building and villa/fit-out examples, not high-rise operating scale.

Only navigation, view filters, role previews, record reading and report viewing were used. No business document was edited, approved, posted, paid, closed or deleted. The dashboard opened a generated report view. No source-code audit, data migration, accounting reconciliation, approval-transition test, load test or actual-user study was performed.

Accessibility observations are risks to verify: truncation, small/dense text, colour-dependent cues and tab-state semantics. Keyboard/focus, screen-reader behaviour, measured contrast and error recovery were not tested. Screenshot blank capture margins are omitted only in PDF placement; visible content is unchanged. Early clipped capture E07 is excluded from visual conclusions in favour of E33.